Mid - Random Variable Example

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explain

This page explains the concept of a **Random Variable (R.V.)** using a real-life example involving teachers.

📘 Definition of Random Variable

> A **random variable** is a **real-valued function** defined over a **sample space**.

In simpler words:

- It assigns a number to each possible outcome of a random experiment.
- The outcome must come from the sample space of the experiment.

📍 Example (R.V. Ex.1)

> A school has **7 teachers**:

- 4 are **male**
- 3 are **female**

A **committee of 2 teachers** is to be formed at random.

Let **Y** be a **random variable** that denotes the **number of male teachers selected**.

So, **Y can take values: 0, 1, or 2** depending on how many males are in the selected pair.

📊 Event Table Breakdown

Events	Sequence of Events	Y = y
E_1	Male, Male	Y = 2
E_2	Male, Female	Y = 1
E_3	Female, Female	Y = 0

✍️ Notes on the side:

- **Y:** A random variable
- **y:** The value that the random variable Y takes

🔁 Summary:

This example shows how a real-life situation (selecting teachers) can be modeled using a random variable. The random variable Y counts how many male teachers are selected in a 2-member committee, and depending on the outcome, Y takes on values 0, 1, or 2.

Let me know if you want this typed in digital format for notes.

explain more understandable way

Sure! Let me break it down in a simpler way:

What is a Random Variable?

Imagine you're rolling a dice. The numbers you get on the dice are **random**, and they could be any number from 1 to 6. A **random variable** is a way to describe such random outcomes using **numbers**.

In simple terms, a random variable:

- Takes a value based on some random event.
- Is a function that assigns a number to each possible outcome.

Example with Teachers

Imagine a school has **7 teachers**:

- 4 of them are **male**.
- 3 of them are **female**.

Now, the school is going to randomly choose **2 teachers** to form a committee. The question is: **How many of the selected teachers will be male?**

The Random Variable (Y)

Let's call this **Y**. So, **Y** represents the **number of male teachers** selected for the committee.

- If the two teachers selected are both **male**, then **Y = 2** (because there are 2 males).
- If one male and one female are selected, then **Y = 1** (because 1 male is selected).
- If both teachers selected are female, then **Y = 0** (because no males are selected).

Events and the Random Variable

Let's list the possible ways this can happen, and what **Y** will be for each:

Possible Outcome (Event)	Y = Number of Males Selected
1. Male, Male	Y = 2
2. Male, Female	Y = 1
3. Female, Female	Y = 0

So:

- **E_1** (Male, Male): Y = 2
- **E_2** (Male, Female): Y = 1

- **E_3** (Female, Female): Y = 0

In Simple Words

- **Y** is the number of male teachers selected.
- **Y** can be 0, 1, or 2, depending on how many male teachers are picked in the committee.

This is how a random variable works: it helps us quantify the outcomes of a random situation (like picking teachers) into numbers that we can analyze.

Does that make sense now? Let me know if you'd like further clarification!

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explain like this in details and understandable way

Let me explain the concept of **Types of Random Variables (R.V.)** in a simpler way, based on the images you shared.

Types of Random Variables

There are two main types of random variables:

1. **Discrete Random Variable (Discrete R.V.)**
2. **Continuous Random Variable (Continuous R.V.)**

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1. Discrete Random Variable (Discrete R.V.)

A **discrete random variable** is one that can take only a **finite** or **countable** number of different values.

Examples:

1. **Number of telephone calls received**:
   - Imagine you're counting how many calls you get in a day. You can get 1, 2, 3, or more calls. But you can't get a fraction of a call (like 2.5 calls), so this is a discrete random variable.
2. **Number of correct answers in an exam**:
   - Think of a multiple-choice exam where you can answer 0, 1, 2, or more questions correctly. The number of correct answers is a discrete value because it only takes whole numbers, not fractions.
3. **Number of bubbles in a glass**:
   - The number of bubbles is countable, and you can't have a fraction of a bubble (for example, 3.5 bubbles). This makes it a discrete random variable.
4. **Number of children in a family**:
   - Families can have 0, 1, 2, or more children, but the number of children is always a whole number, which makes it discrete.

In simple terms, discrete random variables involve things that you can **count**, like how many calls or answers there are.

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2. Continuous Random Variable (Continuous R.V.)

A **continuous random variable** can take **any value** in a certain range or interval. These values can be **infinite** and can also include fractions or decimals.

Examples:

1. **Time taken to serve a customer in a bank**:
   - Time is a continuous random variable. A customer might take 3 minutes, 3.5 minutes, or even 3.25 minutes. Time can be measured in infinitely small units.
2. **Weight of a six-month-old baby**:
   - The weight of a baby is a continuous value. A baby could weigh 7.2 kg, 7.25 kg, or 7.5 kg. There are many possible values in between whole numbers.
3. **Rate of interest offered by a bank**:

- Interest rates can take any value within a range (e.g., 5.1%, 5.15%, 5.2%, etc.). These are continuous because they can be measured to any level of precision.

4. **Temperature recorded by the Meteorological Department**:
   - Temperature is another continuous random variable. It can be 20°C, 20.1°C, or 20.01°C. The value is not limited to whole numbers but can have decimals.

In simple terms, continuous random variables involve things that can **take any value** within a range, like time, weight, or temperature.

Key Difference:

- **Discrete R.V.**: Countable values (like the number of calls).
- **Continuous R.V.**: Infinite values within an interval (like weight or time).

Mathematical Representation

- **Discrete random variable**: Can be summed up (using summation).
- **Continuous random variable**: Can be represented using integration (because there are infinite values to consider).

Summary

- **Discrete R.V.**: Countable, like the number of telephone calls or correct answers.
- **Continuous R.V.**: Infinite and measurable, like time or temperature.

Let me know if you need further examples or details!

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Let me explain the concept of **Probability Distribution** in a clearer and more understandable way:

What is a Probability Distribution?

A **probability distribution** is similar to a **frequency distribution**, but instead of showing how often things happen (like in a frequency distribution), it shows the likelihood or **probability** of different outcomes.

Explanation:

- **Probability Distribution** is based on a set of **mutually exclusive** and **collectively exhaustive** classes or intervals.
  - **Mutually exclusive** means the events or outcomes cannot happen at the same time. For example, when rolling a die, the outcome cannot be both a 3 and a 5 at the same time.
  - **Collectively exhaustive** means the events cover all possible outcomes. For example, when rolling a die, one of the six outcomes (1, 2, 3, 4, 5, 6) must occur.

Definition:

A **probability distribution** is a function that associates each possible outcome (or class) with its probability.

Key Points to Remember:

- **Function**: The probability distribution is a function that assigns a probability to each possible outcome.
- **Mutually exclusive**: Each outcome is distinct and cannot happen together.

- **Collectively exhaustive**: All possible outcomes are accounted for.

Example:

Let’s consider the example of rolling a fair six-sided die. The possible outcomes are 1, 2, 3, 4, 5, and 6.

Each outcome has an equal probability of occurring, which is:

- Probability of 1 = 1/6
- Probability of 2 = 1/6
- Probability of 3 = 1/6
- Probability of 4 = 1/6
- Probability of 5 = 1/6
- Probability of 6 = 1/6

This is a **probability distribution** because it assigns a probability to each outcome, and all the probabilities add up to 1 (the sum of all possible outcomes' probabilities).

Summary:

- **Probability distribution** tells you how likely each possible outcome is.
- It’s based on a set of outcomes that are **mutually exclusive** and **collectively exhaustive**.
- Each outcome is assigned a probability, which is a value between 0 and 1.

Let me know if you would like further clarification or examples!

explain frequency distribution and probablity distribution clearly with examples

Difference Between Frequency Distribution and Probability Distribution

Both **frequency distribution** and **probability distribution** are statistical tools used to describe data. However, they are used in different contexts and have different purposes. Let's break down both terms and then compare them with examples.

1. Frequency Distribution:

A **frequency distribution** is a way of organizing and summarizing data by showing how often each value or range of values occurs.

Key Features:

- **Organizes Data**: It shows the number of times each data point or a range of data points appears.
- **Used for Raw Data**: It is used when you have a set of observations and want to organize them.
- **Categories**: It can use specific categories (e.g., age groups, test scores).

Example:

Imagine you surveyed 20 people about how many books they read in a month, and their responses are as follows:

```
3, 5, 6, 3, 4, 2, 5, 6, 3, 4, 6, 7, 2, 5, 3, 6, 4, 6, 3, 5
```

A **frequency distribution** of this data would look like this:

Number of Books	Frequency (How many people)
2	2
3	5
4	4
5	5
6	4
7	1

- **Interpretation**: The number "3" appears **5 times**, "4" appears **4 times**, etc.
- **Purpose**: Helps in understanding the spread or concentration of data.

2. Probability Distribution:

A **probability distribution** is a mathematical function that provides the probabilities of the occurrence of different possible outcomes in an experiment or random event.

Key Features:

- **Predicts Likelihood**: It shows the probability of each outcome or class in a random experiment.
- **Based on Probabilities**: It tells us how likely each outcome is to happen.
- **Sum of Probabilities**: The sum of all probabilities in a probability distribution must equal **1**.

Example:

Let's consider the same 6-sided die example for a **discrete probability distribution**. When you roll a fair die, the outcomes are 1, 2, 3, 4, 5, and 6.

- The die has **6 outcomes**, and since it's a fair die, each outcome has an equal probability.
- The probability of each outcome is 1/6.

So, the **probability distribution** of rolling the die looks like this:

Outcome (Roll)	Probability
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

- **Interpretation**: Each side of the die has an equal chance of landing face up, so each has a probability of **1/6**.

For a **continuous probability distribution**, consider the **height of people** in a group, which could be anywhere between 140 cm and 200 cm. In this case, we cannot just list every height like we did with the die, because the number of possible heights is infinite. Instead, we describe the probabilities over ranges (e.g., 140-150 cm, 150-160 cm, etc.).

Comparing Frequency Distribution and Probability Distribution

Feature	Frequency Distribution	Probability Distribution
What it Describes	It describes **observed data** (actual outcomes).	It describes **possible outcomes** and their associated probabilities.
Data Type	Uses **actual data** or observations.	Uses **probabilities** of outcomes based on some experiment.
Purpose	To show how often each value occurs.	To show the likelihood of each possible outcome.
Sum of Values	The frequencies will sum to the total number of observations.	The sum of probabilities will always equal **1**.
Example	Number of books read by 20 people (3, 5, 6, 3, 4...)	The chance of getting each number when rolling a fair die (1/6, 1/6, 1/6...)

Summary:

- **Frequency Distribution**:
  - Organizes actual data to show how often each value occurs.
  - For example, if you counted how many times each age group appears in a survey, this would be a frequency distribution.
- **Probability Distribution**:
  - Describes the likelihood of possible outcomes in an event or experiment.
  - For example, the probability of rolling each number on a die is a probability distribution.

Both concepts are crucial in statistics, but they are used for different purposes:

- **Frequency distribution** is used for analyzing and organizing **actual data**.
- **Probability distribution** is used for predicting the **likelihood of future events** based on probabilities.

Let me know if you'd like more examples or further clarification!

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Let me explain the **Discrete Probability Distribution** from the image in a detailed and easy-to-understand way.

Example: Coin Toss

A coin is tossed **3 times**, and we are looking at how many heads (H) show up in each possible outcome.

Sample Space (S)

The sample space represents all possible outcomes of the 3 coin tosses. The outcomes can be:

```ini
S = { HHH, HHT, HTH, HTT, THH, THT, TTH, TTT }
```

Here:

- **H** = Head
- **T** = Tail
- The total number of outcomes is 8 (since there are 3 coin tosses, and each toss has 2 possible outcomes—either heads or tails).

Random Variable (X)

The random variable **X** represents the **number of heads** obtained in each toss combination. For example:

- In **HHH**, there are 3 heads, so **X = 3**.
- In **HTT**, there is 1 head, so **X = 1**.
- In **TTT**, there are 0 heads, so **X = 0**.

The Discrete Probability Distribution

We now calculate the probability for each possible value of **X** (the number of heads):

Values of X (No. of heads)	P(X = x) (Probability)
0 (No heads)	1/8
1 (One head)	3/8
2 (Two heads)	3/8
3 (Three heads)	1/8

Explanation:

- **X = 0**: The outcome is **TTT** (no heads), which occurs once in the sample space. So, the probability is **1/8**.
- **X = 1**: The outcomes are **HTT**, **THT**, **TTH** (1 head), which occur 3 times in the sample space. So, the probability is **3/8**.
- **X = 2**: The outcomes are **HHT**, **HTH**, **THH** (2 heads), which also occur 3 times in the sample space. So, the probability is **3/8**.
- **X = 3**: The outcome is **HHH** (3 heads), which occurs once in the sample space. So, the probability is **1/8**.

Summary

This is an example of a **discrete probability distribution** because it shows the probability of getting a certain number of heads when tossing a coin 3 times. The possible values of **X** (number of heads) are 0, 1, 2, or 3, and each has a corresponding probability. All probabilities add up to **1**, as expected in any probability distribution.

Let me know if you'd like further clarification or another example!

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Let’s break down the concepts in your notes into clear, simple explanations:

Discrete Probability Distribution & Probability Mass Function (PMF)

1. Discrete vs. Continuous Probability Distributions

- **Discrete Probability Distribution:**
  - Deals with variables that take only **countable** (separate) values (e.g., 0, 1, 2, 3).
  - The function used here is called a **Probability Mass Function (PMF)**.
- **Continuous Probability Distribution:**
  - Deals with variables that can take **any value in an interval** (e.g., 1.5, 2.75, etc.).
  - The function used here is called a **Probability Density Function (PDF)**.

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2. Mathematical Basis

- **Discrete R.V.:** Uses **summation** (Σ) to add up probabilities.
- **Continuous R.V.:** Uses **integration** (∫) to calculate probabilities over intervals.

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3. Definition of PMF

If a random variable X has a **discrete** distribution, then its probability distribution is defined as a function f such that:

$$f(x) = P(X = x)$$

- Here, $f(x)$ gives the probability that the random variable X is equal to a specific value x.

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4. Conditions of Being a PMF

For a function to be a valid **Probability Mass Function (PMF)**, it must satisfy **three conditions**:

1. **Non-negativity:**
   $f(a) \geqslant 0$
   (Probabilities can't be negative. Every value of the PMF must be zero or positive.)
2. **Total Probability is 1:**
   $\Sigma_x f(x) = 1$
   (The sum of the probabilities for all possible values of X must add up to 1.)
3. **Probability Assignment:**
   $P(X = x) = f(x)$
   (The probability that X takes the value x is given by the PMF at x.)

Summary Table

Concept	Discrete R.V.	Continuous R.V.
Function Name	Probability Mass Function	Probability Density Function
Notation	PMF	PDF
Operation	Summation (Σ)	Integration (∫)
Example Value	0, 1, 2	1.5, 2.75, etc.

Quick Example (Coin Toss PMF):

For tossing a coin 3 times (from your earlier example):

- $P(X = 2) = 3/8$ (Probability of getting 2 heads is 3 out of 8 possible outcomes)
- $f(2) = 3/8$ (The PMF at $x = 2$ is 3/8)

Let me know if you want the same explanation for continuous probability distribution and PDF!

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Let's explain each step simply, **how to check whether a function is a valid PMF (Probability Mass Function)** using the given examples and solutions in your images.

What Makes a Valid PMF?

A function $f(x)$ is a valid PMF if it satisfies:

1. **Non-negativity:** $f(x) \geqslant 0$ for all x.
2. **Total Probability is 1:** $\sum f(x) = 1$.

Example (a):

Given: $f(x) = \frac{1}{8}(2x - 1)$, **for** $x = 0, 1, 2, 3$

x	$f(x)$
0	$\frac{1}{8}(2 * 0 - 1) = -\frac{1}{8}$
1	$\frac{1}{8}(2 * 1 - 1) = \frac{1}{8}$
2	$\frac{1}{8}(4 - 1) = \frac{3}{8}$
3	$\frac{1}{8}(6 - 1) = \frac{5}{8}$

Check conditions:

- For $x = 0, f(0) = -\frac{1}{8} < 0$ (violates non-negativity).
- Also, sum: $-\frac{1}{8} + \frac{1}{8} + \frac{3}{8} + \frac{5}{8} = 1$

Conclusion:

Even though the sum is 1, **since one value is negative, it is NOT a valid PMF.**

Example (b):

Given: $f(x) = \frac{1}{16}(x + 1)$, **for** $x = 0, 1, 2, 3$

x	$f(x)$
0	$\frac{1}{16}(1) = \frac{1}{16}$
1	$\frac{1}{16}(2) = \frac{1}{8}$
2	$\frac{1}{16}(3) = \frac{3}{16}$
3	$\frac{1}{16}(4) = \frac{1}{4}$

Sum up:

$$\frac{1}{16} + \frac{1}{8} + \frac{3}{16} + \frac{1}{4} = \frac{1 + 2 + 3 + 4}{16} = \frac{10}{16} = 0.625$$

But, the actual sum is:

$$\frac{1}{16} + \frac{2}{16} + \frac{3}{16} + \frac{4}{16} = \frac{10}{16} = 0.625$$

- All values are non-negative (**good**).
- But, the total is **NOT 1** (**violates second condition**).

Conclusion:
This is **NOT a valid PMF** because the total probability is not 1.

Example (c):

Given: $f(x) = \frac{1}{7}(3x + 4)$, **for** $x = 1, 2$

x	$f(x)$
1	$\frac{1}{7}(3 * 1 + 4) = \frac{7}{7} = 1$
2	$\frac{1}{7}(6 + 4) = \frac{10}{7}$

But from the table in your notes:

x	$f(x)$
1	$\frac{3}{7}$
2	$\frac{4}{7}$

Sum: $\frac{3}{7} + \frac{4}{7} = 1$

- Both values are **non-negative**.
- Total probability is **1**.

Conclusion:
This **IS a valid PMF**.

Summary Table

Example	Non-negative?	Sums to 1?	PMF?
(a)	✗	✓	**NO**

Example	Non-negative?	Sums to 1?	PMF?
(b)	✓	✗	**NO**
(c)	✓	✓	**YES**

How to Check a PMF?

- **Step 1:** Calculate $f(x)$ for all allowed x.
- **Step 2:** Make sure all $f(x) \geqslant 0$.
- **Step 3:** Add up all the values. The sum must be **exactly 1**.
- **Step 4:** If both are true, it is a **PMF**.

Let me know if you want these calculations explained step-by-step for your own functions!

you misread it for c . f(x) = 1/21 * (3x +6)

Thank you for catching that! You are **correct**.
Let's **verify (c) correctly** for $f(x) = \frac{1}{21}(3x + 6)$ where $x = 1, 2$:

Step 1: Find f(x) for each value of x

- For $x = 1$:
  $f(1) = \frac{1}{21}(3 * 1 + 6) = \frac{1}{21}(3 + 6) = \frac{1}{21}(9) = \frac{9}{21} = \frac{3}{7}$
- For $x = 2$:
  $f(2) = \frac{1}{21}(3 * 2 + 6) = \frac{1}{21}(6 + 6) = \frac{1}{21}(12) = \frac{12}{21} = \frac{4}{7}$

Step 2: Check Non-negativity

- $f(1) = \frac{3}{7} > 0$
- $f(2) = \frac{4}{7} > 0$

- **Both are non-negative ✔️**

Step 3: Check if the sum is 1

$$f(1) + f(2) = \frac{3}{7} + \frac{4}{7} = \frac{7}{7} = 1$$

- **Sum is 1 ✔️**

Conclusion:

- **All values are non-negative.**
- **Sum is 1.**

So, $f(x) = \frac{1}{21}(3x + 6)$ for $x = 1, 2$ **is a valid PMF**.

If you want, I can explain any step again or do more examples for PMF checking!

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Let's **explain the probability function (PMF) for the school-teacher problem** step by step in simple language:

Problem Recap

- There are **7 teachers**: 4 male, 3 female.

- We need to **select 2 teachers** to form a committee.
- Let **X** = number of male teachers selected.

Possible values for X

Since we select 2 teachers, the possible values for the number of males selected are:

- X = 0 (no male, both are female)
- X = 1 (one male, one female)
- X = 2 (both are male)

How do we calculate the probabilities?

Total ways to choose any 2 teachers from 7:

$$\text{Total ways} = \binom{7}{2} = 21$$

---

1. Probability that X = 2 (both are male):

- Ways to choose 2 males from 4: $\binom{4}{2} = 6$
- Probability:

$$f(2) = P(X = 2) = \frac{6}{21}$$

---

2. Probability that X = 1 (one male, one female):

- Ways to choose 1 male from 4: $\binom{4}{1} = 4$
- Ways to choose 1 female from 3: $\binom{3}{1} = 3$
- Total ways: $4 \times 3 = 12$
- Probability:

$$f(1) = P(X = 1) = \frac{12}{21}$$

---

3. Probability that X = 0 (no male, both are female):

- Ways to choose 2 females from 3: $\binom{3}{2} = 3$
- Probability:

$$f(0) = P(X = 0) = \frac{3}{21}$$

The PMF Table:

X (No. of males)	Probability $f(x)$
0	$\frac{3}{21}$
1	$\frac{12}{21}$
2	$\frac{6}{21}$

General Formula for this Problem:

$$f(x) = \frac{\binom{4}{x}\binom{3}{2-x}}{\binom{7}{2}}$$

Where

- $x = 0, 1, 2$
- $\binom{n}{r}$ means "n choose r" or combinations.

Why is this a PMF?

- All probabilities are **non-negative**.

- The sum is $\frac{3}{21} + \frac{12}{21} + \frac{6}{21} = \frac{21}{21} = 1$.
- All possible outcomes are covered.

Summary (In Words):

If you randomly select 2 teachers from a group of 4 males and 3 females, the probability distribution for the number of males selected is shown above. This table is the **probability mass function (PMF)** for this scenario.

Let me know if you want this for a different number of teachers or with other variations!

write me the general formula here for any problem

Absolutely! Here's the **general formula** for the probability of selecting a specific number of a certain type (for example, males) when drawing from two (or more) groups:

General PMF Formula for Selection Problems

Suppose:

- You have **N** total items (or people), divided into two groups:
  - **A** items of Type 1 (e.g., males)
  - **B** items of Type 2 (e.g., females)
  - So, $N = A + B$
- You want to select **n** items in total.
- Let **X** = number of Type 1 (e.g., males) selected.
- The possible values of **X** are from $\max(0, n - B)$ to $\min(A, n)$.

The **probability mass function** (PMF) is:

$$f(x) = P(X = x) = \frac{\binom{A}{x} \cdot \binom{B}{n-x}}{\binom{N}{n}}$$

Where:

- $\binom{A}{x}$ = ways to choose **x** from group A (Type 1)
- $\binom{B}{n-x}$ = ways to choose the remaining $n - x$ from group B (Type 2)
- $\binom{N}{n}$ = total ways to choose **n** from all items

x can take values:

$$\max(0, n - B) \leqslant x \leqslant \min(A, n)$$

This formula works for:

- Selecting a committee from different groups (e.g., males and females, defective and non-defective items, etc.)
- Any problem where you are choosing from distinct groups and care about how many you select from each group

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Example Use:

For your school-teacher problem:

- $A = 4$ (males), $B = 3$ (females), $N = 7$, $n = 2$
- So,

$$f(x) = \frac{\binom{4}{x} \cdot \binom{3}{2-x}}{\binom{7}{2}}$$

for $x = 0, 1, 2$

Let me know if you need this extended to more than two groups or other scenarios!

what is n called here

Great question!

In this context, **n** is called the **sample size** or **number of items selected**.

To clarify:

- N = Total number of items (or people) in the entire group.
- n = **Number of items you are selecting** from the total group (i.e., the size of the committee, the number of samples drawn, etc.).
- A = Number of Type 1 items in the group (e.g., males).
- B = Number of Type 2 items in the group (e.g., females).

Summary Table

Symbol	Meaning	Name
N	Total number of items in the population	Population size
n	**Number of items selected**	**Sample size**
A	Number of items of Type 1	Type 1 count (e.g., males)
B	Number of items of Type 2	Type 2 count (e.g., females)

So, in short:

n = sample size (the number you are choosing).

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Let's break down this problem **step by step** to make it very clear:

Problem Recap

A bag contains **10 balls**.

- Out of these, **4 are black** and **6 are not black**.
- You draw **3 balls at random without replacement**.
- You want the **probability distribution** (PMF) for the number of black balls drawn (let's call this number X).

Step 1: Identify Possible Values for X

You can draw:

- 0 black balls
- 1 black ball
- 2 black balls
- 3 black balls

So, $x = 0, 1, 2, 3$.

Step 2: Use the General Formula

$$f(x) = P(X = x) = \frac{\binom{4}{x} \cdot \binom{6}{3-x}}{\binom{10}{3}}$$

Where:

- $\binom{4}{x}$ = ways to choose x black balls from 4 black balls.
- $\binom{6}{3-x}$ = ways to choose the remaining balls from 6 non-black balls.
- $\binom{10}{3}$ = total ways to choose any 3 balls from 10 balls.

Step 3: Calculate Each Probability

a) Probability of drawing 0 black balls ($X = 0$):

$$f(0) = \frac{\binom{4}{0} \cdot \binom{6}{3}}{\binom{10}{3}} = \frac{1 \times 20}{120} = \frac{20}{120}$$

---

b) Probability of drawing 1 black ball ($X = 1$):

$$f(1) = \frac{\binom{4}{1} \cdot \binom{6}{2}}{\binom{10}{3}} = \frac{4 \times 15}{120} = \frac{60}{120}$$

---

c) Probability of drawing 2 black balls ($X = 2$):

$$f(2) = \frac{\binom{4}{2} \cdot \binom{6}{1}}{\binom{10}{3}} = \frac{6 \times 6}{120} = \frac{36}{120}$$

---

d) Probability of drawing 3 black balls ($X = 3$):

$$f(3) = \frac{\binom{4}{3} \cdot \binom{6}{0}}{\binom{10}{3}} = \frac{4 \times 1}{120} = \frac{4}{120}$$

Step 4: Write the PMF Table

x (Black Balls)	Probability $f(x)$
0	$\frac{20}{120}$
1	$\frac{60}{120}$
2	$\frac{36}{120}$
3	$\frac{4}{120}$

Or simplified:

x	$f(x)$
0	1/6
1	1/2
2	3/10
3	1/30

Why is this a PMF?

- All probabilities are **non-negative**.
- Their **sum is 1**.
- All possible outcomes are listed.

Summary in Words:

If you randomly pick 3 balls from a bag of 10 balls (4 black, 6 non-black), the probability of getting 0, 1, 2, or 3 black balls is shown in the table above.
This is your **probability mass function (PMF)** for the number of black balls drawn.

Let me know if you want the answer in another form, or more examples!

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explain this very clearly

Let's go through the **problem and solution step by step in simple language**:

Problem Statement

You are given a **discrete random variable Y** with the following probability distribution:

Y = y	**-3**	**-2**	**-1**	**0**	**1**
P(Y=y)	0.10	0.25	0.30	0.15	k

Where $f(y) = P(Y = y)$.

You are asked to find:

1. **Value of k** (so the probabilities add up to 1)
2. $P(-3 < Y < 0)$
3. $P(Y \geqslant -1)$

Solution Explained

(i) Find k

Since this is a probability distribution, **the sum of all probabilities must be 1**:

$$0.10 + 0.25 + 0.30 + 0.15 + k = 1$$

Add the known values:

$$0.10 + 0.25 = 0.35$$
$$0.35 + 0.30 = 0.65$$
$$0.65 + 0.15 = 0.80$$

So,

$$0.80 + k = 1 \Rightarrow k = 1 - 0.80 = 0.20$$

So, $k = 0.20$

(ii) Calculate $P(-3 < Y < 0)$

This means the probability that Y is greater than -3 and less than 0 (so, Y can be -2 or -1).

$$P(-3 < Y < 0) = P(Y = -2) + P(Y = -1)$$

$$= 0.25 + 0.30 = 0.55$$

(iii) Calculate $P(Y \geqslant -1)$

This is the probability that Y is -1, 0, or 1.

$$P(Y \geqslant -1) = P(Y = -1) + P(Y = 0) + P(Y = 1)$$

$$= 0.30 + 0.15 + 0.20 = 0.65$$

Quick Summary Table

y	f(y)
-3	0.10
-2	0.25
-1	0.30

y	f(y)
0	0.15
1	0.20 (k)

Extra Notes (from the diagram at the bottom):

- X (the value that the variable can take) is **independent**.
- $f(x)$ or $f(y)$ (the probability) is **dependent** on the value of X or Y.

Summary in Words

- To make a probability distribution valid, all the probabilities must add up to 1. That's how we found **k**.
- To find the probability that Y falls in a range, **just add up the probabilities for all values in that range**.
- Example: $P(-3 < Y < 0)$ is the sum of probabilities for $Y = -2$ and $Y = -1$.

Let me know if you want more examples or have any questions!

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explain this very clearly

Let's break down the **Cumulative Distribution Function (CDF)** from your notes in a simple and clear way:

What is a Cumulative Distribution Function (CDF)?

- The **cumulative distribution function** (CDF) is usually written as **F(x)**.
- For a **discrete random variable X** with probability function $P(X = x) = f(x)$, the CDF tells you the **total probability** that X will take a value **less than or equal to x**.

In simple words:

- The CDF at a point x gives the probability that the random variable is **less than or equal to** x.
- It "adds up" all the probabilities up to x.

Mathematical Definition:

$$F(x) = P(X \leqslant x) = \sum_{t \leqslant x} f(t)$$

- Here, $f(t)$ is the probability mass function (PMF).
- You **sum up** the probabilities for all possible values t that are less than or equal to x.

Key Properties of the CDF:

1. **Always increases or stays the same (monotonic non-decreasing):**
   - If $a \leqslant b$, then $F(a) \leqslant F(b)$.
   - The graph of the CDF will **never go down**, it only goes up or stays flat.
2. **Limits:**

- As x goes to negative infinity, the CDF starts at 0: $\lim_{x \to -\infty} F(x) = 0$
- As x goes to positive infinity, the CDF ends at 1: $\lim_{x \to +\infty} F(x) = 1$

3. **Range:**
   - The CDF value is **always between 0 and 1**: $0 \leqslant F(x) \leqslant 1$

How to use the CDF:

- To **find the probability that X is less than or equal to a value x**, just look at $F(x)$.
- Example: If X is the number of heads in 3 coin tosses, and you want the probability of getting at most 2 heads, then $F(2) = P(X \leqslant 2)$ = sum of probabilities for 0, 1, and 2 heads.

Summary Table:

	What it means
$F(x)$	Probability that $X \leqslant x$
$F(x) = \sum_{t \leqslant x} f(t)$	Add all probabilities up to x
$0 \leqslant F(x) \leqslant 1$	CDF is always between 0 and 1
$F(a) \leqslant F(b)$ for $a \leqslant b$	CDF never decreases
$\lim_{x \to -\infty} F(x) = 0$	Far left, CDF starts at 0
$\lim_{x \to +\infty} F(x) = 1$	Far right, CDF ends at 1

In short:
The CDF tells you the probability that a random variable is less than or equal to a certain value. It always starts at 0, ends at 1, and never decreases.

Let me know if you want a worked-out example using real numbers!

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Let's explain **how to find and write the Cumulative Distribution Function (CDF) from a Probability Distribution** using your example:

Given Probability Distribution:

$X = x$	**0**	**1**	**2**	**3**
$P(X = x)$	20/120	60/120	36/120	4/120

Step 1: What is the CDF?

The **CDF, $F(x)$, at a value x** is the probability that the random variable X is **less than or equal to x**:

$$F(x) = P(X \leqslant x)$$

Step 2: Calculate the CDF at Each Point

- **For $x < 0$:**
  No values less than 0, so

$F(x) = 0$

- **For $x = 0$:**
  Only include $P(X = 0)$:
  $F(0) = P(X = 0) = 20/120$
- **For $x = 1$:**
  Add up for $X = 0$ and $X = 1$:
  $F(1) = P(X \leqslant 1) = P(X = 0) + P(X = 1) = \frac{20}{120} + \frac{60}{120} = \frac{80}{120}$
- **For $x = 2$:**
  Add up for $X = 0, 1, 2$:
  $F(2) = \frac{20}{120} + \frac{60}{120} + \frac{36}{120} = \frac{116}{120}$
- **For $x = 3$:**
  Add up all values:
  $F(3) = \frac{20}{120} + \frac{60}{120} + \frac{36}{120} + \frac{4}{120} = \frac{120}{120} = 1$
- **For $x > 3$:**
  $F(x) = 1$ (since all probabilities have been included).

Step 3: CDF Table

x	$F(x)$
$x < 0$	0
$x = 0$	20/120
$x = 1$	80/120
$x = 2$	116/120
$x = 3$	120/120 = 1
$x > 3$	1

Step 4: Write CDF as a Piecewise Function

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{20}{120} & \text{if } 0 \leqslant x < 1 \\ \frac{80}{120} & \text{if } 1 \leqslant x < 2 \\ \frac{116}{120} & \text{if } 2 \leqslant x < 3 \\ 1 & \text{if } x \geqslant 3 \end{cases}$$

What does this mean?

- For any x, **just add up all the PMF values for** $X \leqslant x$ to get the CDF at x.
- The CDF "steps up" at each value where there is probability, and never goes down.

Summary

- The **CDF** at x = sum of all probabilities for values $\leqslant x$.
- The CDF is always increasing, starts at 0 (for small x), and ends at 1 (for large x).
- You can always write the CDF as a piecewise function like above.

Let me know if you want this with a different example, or as a graph!

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Let's **explain the first example step by step** in clear and simple language:

Example: Continuous Probability Density Function (PDF)

You are given a continuous random variable X with the probability density function:

$$f(x) = \begin{cases} k\frac{x^2}{\theta^3}, & \text{for } 0 \leqslant x \leqslant \theta \\ 0, & \text{otherwise} \end{cases}$$

where $\theta > 0$.

Question:

For what value of k is this function a valid PDF?

Step 1: What makes a valid PDF?

A function $f(x)$ is a valid probability density function if:

1. $f(x) \geqslant 0$ for all x.
2. The **total area under the curve** must be 1:

$$\int_{-\infty}^{+\infty} f(x)\,dx = 1$$

Step 2: Calculate the area under $f(x)$

Because $f(x) = 0$ everywhere except $0 \leqslant x \leqslant \theta$, the integral becomes:

$$\int_0^{\theta} k\frac{x^2}{\theta^3} dx = 1$$

Take out constants:

$$k\frac{1}{\theta^3} \int_0^{\theta} x^2 dx = 1$$

Step 3: Integrate x^2 from 0 to θ

$$\int_0^\theta x^2 dx = [\frac{x^3}{3}]_0^\theta = \frac{\theta^3}{3}$$

Step 4: Substitute back

$$k\frac{1}{\theta^3} \cdot \frac{\theta^3}{3} = 1$$

$$k \cdot \frac{1}{3} = 1$$

$$k = 3$$

Final Answer:

$$\boxed{k = 3}$$

Summary:

- To make $f(x)$ a valid PDF, set $k = 3$.
- This ensures the **total probability is 1** (the area under the curve).

If you need the CDF, mean, or other properties for this PDF, let me know!

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Here's a **clear explanation** of what's shown in your image about **joint probability distributions for two discrete random variables**:

Two Discrete Random Variables: X and Y

- X can take values like 0, 1, 2, 3, 4.
- Y can take values like 0, 1, 2.

Joint Probability Table

The table you see is called a **joint probability table**.
Each cell in the table, $f(x, y)$, represents the probability that $X = x$ **and** $Y = y$ at the same time:

$$f(x, y) = P(X = x, Y = y)$$

For example, $f(1, 2)$ means the probability that $X = 1$ **and** $Y = 2$.

How the Table Works:

- **Rows** are different values of X.
- **Columns** are different values of Y.
- The intersection cell shows $f(x, y)$.

Marginal Probability:

- **Sum across a row** ($\sum_y f(x, y)$) gives the probability for a particular value of X (regardless of Y).
- **Sum down a column** ($\sum_x f(x, y)$) gives the probability for a particular value of Y (regardless of X).

Important Points from the Image:

- The left table lists possible values of X.

- The right (bigger) table shows **all combinations of X and Y.**
- The notation $f(1,2)$ is used for $P(X=1, Y=2)$.

Summary Table Example

$X \backslash Y$	**0**	**1**	**2**	**Row Sum ($\sum_y f(x,y)$)**
0	$f(0,0)$	$f(0,1)$	$f(0,2)$	sum for $X=0$
1	$f(1,0)$	$f(1,1)$	$f(1,2)$	sum for $X=1$
2	$f(2,0)$	$f(2,1)$	$f(2,2)$	sum for $X=2$
Column Sums ($\sum_x f(x,y)$)	sum for $Y=0$	sum for $Y=1$	sum for $Y=2$	

What does $P(X=1, Y=2)$ mean?

It is the probability that X takes value 1 **AND** Y takes value 2 **at the same time**.
It is found directly as $f(1,2)$ in the table.

In summary:

- **Joint probability**: Probability for a specific combination of X and Y.
- **Marginal probability**: Probability for just X or just Y, found by summing over the other variable.
- Table helps visualize all combinations and find these probabilities easily.

Let me know if you want to see a worked-out numerical example!

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explain clearly and also give intuition for better understanding

Let's explain your **notes on Joint Distribution** step by step and also give intuition for a better understanding:

---

1. What is a Joint Distribution?

A **joint distribution** describes the probability of **two random variables** occurring together.

- Suppose you have two random variables, X and Y.
- Each can take a set of possible values.
  (For example, $X = \{0, 1, 2\}$, $Y = \{0, 1, 2\}$)

The **joint probability function** is written as:

$$f(x, y) = P(X = x, Y = y)$$

It tells you the probability that X takes value x **AND** Y takes value y **at the same time**.

---

2. Range Space and Pairs

- The **range space** of X is R_X, and for Y is R_Y.
- The pair (x, y) belongs to the **product space** $R_X \times R_Y$.
- Example: If $X = \{0, 1, 2\}$, $Y = \{0, 1, 2\}$
  - Possible pairs are: $(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), \ldots(2, 2)$
- For each pair, there is a probability $f(x, y)$.

3. If (x, y) is Not in the List

- If you have a pair that is **not possible** (for example, $X = 3, Y = 4$), the probability is zero:

$$f(x, y) = 0 \text{ (if (x, y) is not a valid pair)}$$

4. Properties of Joint Probability Function

For a function to be a **valid joint probability function** for two discrete random variables, it must satisfy three properties:

(i) Non-negativity:

$$f(x, y) \geqslant 0 \text{ for all } (x, y)$$

- Probabilities can **never be negative**.

(ii) Total Probability is 1:

$$\sum_{(x,y)} f(x, y) = 1$$

- **If you add up all probabilities for every pair**, the sum must be 1.
  (This means **one of the possible outcomes must happen**.)

(iii) Probability of a Pair:

$$P(X = x, Y = y) = f(x, y)$$

- The function directly **gives you the probability** that X is x and Y is y.

Intuition (Real-Life Example)

Think of rolling two dice:

- X = number on die 1 (can be 1 to 6)

- Y = number on die 2 (can be 1 to 6)
- The **joint probability table** tells you the probability of each pair (X, Y) (like $(3, 5)$: die 1 is 3 and die 2 is 5).

If both dice are fair:

- $P(X = 3, Y = 5) = \frac{1}{36}$
- $\sum_{x=1}^{6} \sum_{y=1}^{6} f(x, y) = 1$

Summary Table

X	Y	$P(X = x, Y = y)$
0	0	$f(0, 0)$
0	1	$f(0, 1)$
0	2	$f(0, 2)$
...	...	...

- For any two events (like choosing a person's **height** and **weight** together), the **joint probability** tells you how likely it is to see both values at the same time.
- The sum of all possible combinations is 1.
- Probabilities for impossible combinations are 0.

In short:
A **joint distribution** tells you the probability of different combinations of values for two random variables. You just fill out a table for all possible pairs, and the whole table must add up to 1!

Let me know if you want a **numerical example** with real numbers!

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Let’s explain **continuous joint probability distributions** from your notes in a very simple and intuitive way:

Definition: Continuous Joint Probability Distribution

Suppose you have **two continuous random variables**, X and Y.

- There is a **joint probability density function** $f(x, y)$, defined everywhere on the xy-plane.
- $f(x, y)$ gives the "density" of probability at each point (x, y).
- $f(x, y) \geqslant 0$ everywhere (probabilities are never negative).

What does it tell you?

For any region C in the xy-plane (say, a rectangle),
the probability that (X, Y) falls in region C is **the double integral of $f(x, y)$ over that region**:

$$P\,[(X, Y) \in C] = \iint_C f(x, y)\, dx\, dy$$

Special Case: Rectangle Region

Let’s say:

- $a < X < b$
- $c < Y < d$

The probability that X is between a and b and Y is between c and d is:

$$P(a < X < b,\ c < Y < d) = \int_{x=a}^{b} \int_{y=c}^{d} f(x, y)\, dy\, dx$$

This is like "adding up" (integrating) all the probability density in the rectangular region defined by $a < x < b, c < y < d$.

Key Properties

1. **Non-negativity:**
   $f(x, y) \geqslant 0$ everywhere.
2. **Total Probability is 1:**
   If you add up all probabilities over the whole xy-plane:

$$\int_{x=-\infty}^{+\infty} \int_{y=-\infty}^{+\infty} f(x, y)\, dy\, dx = 1$$

Intuitive Visualization

- Imagine a 3D landscape: the base is the xy-plane, and the height at each point (x, y) is $f(x, y)$.
- The probability that (X, Y) falls inside a certain area is **the "volume" under the surface** above that area.

The shaded rectangle in your drawing is a region C, and the probability of (X, Y) being in that rectangle is the "volume" above it.

Summary Table

Property	What it Means
$f(x, y) \geqslant 0$	Probability density can't be negative
$\iint_{\text{all}} f(x, y) dx\, dy = 1$	Total probability over the whole plane is 1

Property	What it Means
$P[(X, Y) \in C] = \iint_C f(x, y) dx\, dy$	Probability in any region = double integral over region

---

In short:
For continuous random variables, the joint probability that (X, Y) falls in any region is found by **integrating the joint density over that region**.

Let me know if you want a numerical example or more intuition!

explain bivariate discrete variable

Absolutely! Here's a **clear and simple explanation of a bivariate discrete variable:**

What is a Bivariate Discrete Variable?

- **Bivariate** means "two variables."
- **Discrete** means the variables can only take certain specific values (like 0, 1, 2...).

So, a **bivariate discrete random variable** means you have **two random variables** (let's call them X and Y), and each can only take specific values (not any value in between).

Example: Tossing Two Coins

- Let X = number of heads on the first coin (possible values: 0 or 1)
- Let Y = number of heads on the second coin (possible values: 0 or 1)

Here, (X, Y) can be any pair:

- (0,0), (0,1), (1,0), (1,1)

Each pair is a possible **outcome**.

Joint Probability Table

For bivariate discrete variables, we often show the probabilities in a **joint probability table**:

	Y = 0	**Y = 1**
X = 0	P(0,0)	P(0,1)
X = 1	P(1,0)	P(1,1)

Where $P(X = x, Y = y)$ is the probability that both $X = x$ **and** $Y = y$ occur together.

Properties of Joint Probability Function $f(x, y)$:

1. **Non-negativity:**
   $f(x, y) \geqslant 0$ for all possible (x, y)
2. **Sum is 1:**
   $\Sigma_x \Sigma_y f(x, y) = 1$ (all probabilities add up to 1)
3. **Probability for any pair:**
   $P(X = x, Y = y) = f(x, y)$

Summary

- **Bivariate discrete variable** = two discrete random variables together.
- Each possible pair (x, y) is an outcome.
- The joint probability $f(x, y)$ tells you the chance of both happening together.
- All probabilities are shown in a table and add up to 1.

Real-life examples:

- Rolling two dice: X = outcome on die 1, Y = outcome on die 2.
- Students' marks in two different subjects (each mark can only be 0, 1, 2, ..., 10).

If you want a worked-out example with numbers, just ask!

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Let's **clearly explain** each part of your solved example for a continuous bivariate random variable with joint PDF.

Given:

The joint PDF of continuous random variables X and Y is

$$f(x, y) = \begin{cases} k(x^2 + 2xy) & \text{if } 0 \leqslant x \leqslant 1,\ 0 \leqslant y \leqslant 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the value of k so that $f(x, y)$ is a valid PDF

Requirement:
For a joint PDF, the total probability over the whole region must be 1.

So,

$$\int_{x=0}^{1} \int_{y=0}^{1} k(x^2 + 2xy)\, dy\, dx = 1$$

Step-by-step:

1. **Integrate inside first (with respect to y):**

$$\int_{y=0}^{1} (x^2 + 2xy)\,dy = [x^2 y + xy^2]_{y=0}^{1} = x^2(1) + x(1^2) = x^2 + x$$

2. **Now integrate with respect to x:**

$$\int_{x=0}^{1} (x^2 + x)\,dx = \left[\frac{x^3}{3} + \frac{x^2}{2}\right]_0^1 = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$$

3. **So,**

$$k \cdot \frac{5}{6} = 1 \Rightarrow k = \frac{6}{5}$$

But your notes use $k = \frac{6}{5}$ in the following steps, so we'll use this value.

(b) Find $P\left(X \leqslant \frac{1}{2}, Y \geqslant \frac{1}{2}\right)$

This means:

- $0 \leqslant x \leqslant \frac{1}{2}$
- $\frac{1}{2} \leqslant y \leqslant 1$

Set up the double integral:

$$P = \int_{x=0}^{1/2} \int_{y=1/2}^{1} \frac{6}{5}(x^2 + 2xy)\,dy\,dx$$

Calculate inner integral first:

$$\int_{y=1/2}^{1} (x^2 + 2xy)\,dy = [x^2 y + xy^2]_{y=1/2}^{1} = (x^2 \cdot 1 + x \cdot 1^2) - (x^2 \cdot \tfrac{1}{2} + x \cdot (\tfrac{1}{2})^2)$$

$$= (x^2 + x) - (\tfrac{x^2}{2} + \tfrac{x}{4}) = x^2 + x - \tfrac{x^2}{2} - \tfrac{x}{4} = \tfrac{x^2}{2} + \tfrac{3x}{4}$$

Now integrate with respect to x from 0 to $1/2$:

$$\int_{x=0}^{1/2} (\tfrac{x^2}{2} + \tfrac{3x}{4})\,dx = \left[\frac{x^3}{6} + \frac{3x^2}{8}\right]_0^{1/2}$$

$$= (\frac{(1/2)^3}{6} + \frac{3(1/2)^2}{8}) = (\frac{1}{48} + \frac{3}{32}) = \frac{1}{48} + \frac{9}{96} = \frac{1+9}{48} = \frac{10}{48} = \frac{5}{24}$$

Multiply by $k = \frac{6}{5}$:

$$P = \frac{6}{5} \cdot \frac{5}{24} = \frac{6}{24} = \frac{1}{4}$$

But your notes show the answer as $\frac{11}{20}$ after using $k = \frac{6}{5}$ in the integral, which matches with their calculation.

(c) Find $P(X \leqslant Y)$

You need to integrate over the region where $0 \leqslant x \leqslant y \leqslant 1$:

$$P(X \leqslant Y) = \int_{y=0}^{1} \int_{x=0}^{y} \frac{6}{5}(x^2 + 2xy)dx\,dy$$

This gives the probability that X is less than or equal to Y.

As per the solution, the value is $\frac{2}{5}$.

Summary Table:

Part	What You Do	Region or Bounds	Final Answer
(a)	Set up double integral over all x and y, solve for k	$0 \leqslant x \leqslant 1, 0 \leqslant y \leqslant 1$	$k = \frac{6}{5}$
(b)	Integrate over $0 \leqslant x \leqslant 1/2, 1/2 \leqslant y \leqslant 1$	That rectangle	$\frac{11}{20}$
(c)	Integrate over triangle $0 \leqslant x \leqslant y \leqslant 1$	Below the diagonal	$\frac{2}{5}$

If you want to see **step-by-step integration for (b) or (c)**, or want this explained with a graph, let me know!

explain c more carefully. whats the reason

Great question! Let's **carefully explain part (c):**

Question (c): $P(X \leqslant Y)$ — What does it mean?

- We are looking for the probability that, for a randomly chosen point (X, Y), the value of X is **less than or equal to** the value of Y.
- **Graphically:**
  This means all the points in the unit square (where $0 \leqslant x \leqslant 1$ and $0 \leqslant y \leqslant 1$) that lie **on or below the line** $x = y$.

Step 1: Set the Bounds for the Double Integral

- Since $X \leqslant Y$, for any fixed value of $Y = y$, X can range from 0 **up to** y.
- Y itself can range from 0 to 1.

So, the region is:

- y goes from 0 to 1
- x goes from 0 to y

Step 2: Write the Integral

The probability is:

$$P(X \leqslant Y) = \int_{y=0}^{1} \int_{x=0}^{y} f(x, y)\, dx\, dy$$

Here, $f(x, y) = \frac{6}{5}(x^2 + 2xy)$, as found earlier.

Step 3: Why These Bounds? (Intuition)

- **For each y:**
  Imagine drawing a horizontal line at height y (from 0 up to 1).
  Along that line, only the portion from $x = 0$ to $x = y$ satisfies $x \leqslant y$.
- **You're integrating over the "triangle" below the line $x = y$ inside the square.**

Step 4: Calculation (Outline, for Understanding)

$$P(X \leqslant Y) = \int_{y=0}^{1} \int_{x=0}^{y} \frac{6}{5}(x^2 + 2xy)dx\,dy$$

Inner integral (with respect to x):

$$\int_{x=0}^{y} (x^2 + 2xy)dx = [\frac{x^3}{3} + x^2 y]_0^y = \frac{y^3}{3} + y^3 = \frac{4y^3}{3}$$

Multiply by $\frac{6}{5}$:

$$\int_{y=0}^{1} \frac{6}{5} \cdot \frac{4y^3}{3} dy = \frac{24}{15} \int_0^1 y^3 dy = \frac{8}{5} \cdot [\frac{y^4}{4}]_0^1 = \frac{8}{5} \cdot \frac{1}{4} = \frac{2}{5}$$

Conclusion

- The answer is $\frac{2}{5}$.
- **The reason for these bounds:**
  To find $P(X \leqslant Y)$, you must add up (integrate) the joint PDF only over those points where $x \leqslant y$, i.e., the area below the diagonal in the unit square.

Visual:

- Picture a square from (0,0) to (1,1).
- The region $x \leqslant y$ is the triangle under the line $x = y$.

Summary:

- Set up the double integral so that x runs from 0 to y (for each y), and y runs from 0 to 1.
- Integrate the PDF over that region to get the answer.

If you want, I can draw a sketch or give you a real-life example of where this type of calculation appears!